

Ryu Hyunseung

I build web apps — and the languages and runtimes they run on.

I move between frontend work and systems programming. Using an abstraction layer and building one teach different things, and I want both.

South Korea ryu@rhseung.me github.com/rhseung rhseung.me

Experience

Frontend developer intern

2025.09 – Present

Audio Intelligence Technology & Research Lab (AiTeR), School of EECS, GIST

Taking part in experiments and research at the audio intelligence lab.

aiter.gist.ac.kr/ ↗

Education

Undeclared, GIST College

2025.03 – Present

Gwangju Institute of Science and Technology (GIST)

Freshman representative at the 2025 matriculation ceremony

Natural Sciences

2022.03 – 2025.02

Gwangju Science Academy for the Gifted

Computer science GPA 4.285 / 4.3

Selected projects

rhseung.me

2026.08

A personal site built on Astro islands. This page is the result.

Campass

2024.11 – 2024.11

A companion service that gathers everything you need for campus life in one place.

Table-tennis robot

2024.10 – 2025.01

Computer vision tracks the ball in real time and predicts the strike point to drive a four-axis linear robot.

NEAT

2024.09

Evolving network topology itself with NEAT (NeuroEvolution of Augmenting Topologies).

Glance

2024.06

A Minecraft interface mod bundling a set of quality-of-life tweaks.

Fliggle

2024.05 – 2024.07

A social app built for practice, with both the frontend and the backend written from scratch.

IdP	2024.03 – 2024.08
GIST's single sign-on system. OAuth 2.0 and OpenID Connect let one account log into every campus service, and external ones too.	
Particles	2024.03 – 2024.11
A constraint-based 2D physics engine. Position-based dynamics solve rigid bodies and joints.	
Pattern	2024.02 – 2024.05
Builds regular expressions through a builder API — method chains instead of unreadable expressions.	
DateTime	2024.01 – 2024.04
Parses natural-language expressions like "next Tuesday" into dates.	
siunits	2023.12
A unit-arithmetic library for physics experiments. Dimensionally invalid operations fail at the type level.	
GSA KakaoTalk bot	2023.09 – 2024.02
A KakaoTalk bot for Gwangju Science Academy — meals, timetables and the academic calendar without leaving the chat.	
YOLOv3 from scratch	2023.03 – 2023.11
Reimplemented YOLOv3 from the paper alone — anchor boxes and multi-scale detection included.	
Awards & achievements	
Winner, 2025 Honam Regional Mini Tex-Corps	2025
Ministry of Science and ICT · COMPA Hosted by GIST, Ewha Womans University and Wonkwang University startup centers	
Freshman representative, 2025 matriculation ceremony	2025
Gwangju Institute of Science and Technology (GIST)	
Academic excellence scholarship (science & engineering)	2025
Gwangju Institute of Science and Technology (GIST) Spring 2025	
GS1001 Calculus and Applications — 1st	2025
Gwangju Institute of Science and Technology (GIST) 240 / 240	
GS1401 Computer Programming — 1st	2025
Gwangju Institute of Science and Technology (GIST) 98 / 100	
Participant, AI Creative Convergence Competition	2025
GIST AI Graduate School Table-tennis robot track	

Participant, JUNCTION ASIA 2025 ENTBE	2025
Participant, L:AUNCH AI Career School Root Impact · Krypton X Supported by Google.org	2025
Data Structures & Algorithms — 2nd Gwangju Science Academy for the Gifted Year 3, semester 1	2024
Subject Excellence Award, Computer Science Project Gwangju Science Academy for the Gifted Year 3, semester 1 · 1st	2024
Gold Prize, in-school competition (computer science) Gwangju Science Academy for the Gifted	2023
Subject Excellence Award, Machine Learning & Deep Learning Gwangju Science Academy for the Gifted Year 2, semester 2 · 1st	2023
Subject Excellence Award, Object-Oriented Programming Gwangju Science Academy for the Gifted Year 2, semester 2 · 1st	2023
Subject Excellence Award, Programming & Problem Solving Gwangju Science Academy for the Gifted Year 2, semester 2 · 1st (AP)	2023
Gold Prize, in-school competition (computer science) Gwangju Science Academy for the Gifted Only gold prize among first-year students	2022
Subject Excellence Award, Computer Science I Gwangju Science Academy for the Gifted Year 1, semester 1 · 5th	2022
Subject Excellence Award, Computer Science II Gwangju Science Academy for the Gifted Year 1, semester 2 · 2nd	2022

Skills

Languages

TypeScript

JavaScript

Python

Java

Kotlin

C

C++

Dart

Web

React Next.js Astro TanStack Router TanStack Query React Router Tailwind CSS Vanilla Extract
Styled Components i18next Vite HTML CSS

Backend & data

Node.js Nest.js Flask Prisma MySQL OpenAPI TypeScript

Machine learning

PyTorch TensorFlow Keras scikit-learn OpenCV NumPy Pandas Matplotlib Seaborn Altair
Jupyter

Apps & graphics

Flutter SFML Swing

Tooling

Bun npm Yarn PyPI Figma LaTeX